When prices move together: connectedness among CPI components, wages, and expectations across the inflation distribution and its consequences

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Abstract

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1 Introduction

Inflation targeting has made inflation one of the most closely monitored macroeconomic indicators. When inflation rises sharply, attention shifts from headline inflation alone to the breadth, persistence and propagation of price increases. In 2021 at Jackson Hole, Powell (2021) played down the rise in inflation as narrow and likely temporary, describing it as "largely the product of a relatively narrow group of goods and services that have been directly affected by the pandemic." By 2022, that assessment had changed: Powell (2022) observed that "high inflation has continued to spread through the economy" and, drawing on the lessons of the 1970s, warned that the anticipation of high inflation can become "entrenched in the economic decisionmaking of households and businesses."

Powell's remarks suggest two questions become central for monetary policy when inflation rises. First, are price increases concentrated in a small number of categories, or have they become broad-based across the household expenditure basket? A relative price shock affecting a limited set of goods can often be looked through; a generalised increase across many categories is harder to dismiss (Bryan & Cecchetti, 1994). Second, has inflation become self-reinforcing through second-round effects in wages and inflation expectations? If higher prices raise wage demands and inflation expectations, and those wages and expectations then feed back into prices, inflation may become more persistent and costly to reduce.

[TODO: [lit review here]]

This paper examines whether high inflation changes the connectedness—the strength of statistical association—among prices, wages and inflation expectations in the United States, and whether changes in price, wage and inflations expectation connectedness has consequences for the conduct of monetary policy. Connectedness measures, developed to quantify directional spillovers in financial systems (Diebold & Yilmaz, 2012), are well suited to this question because they provide a direct measure of how movements in one series are associated with movements in others. The closest prior contribution is Stenfors et al. (2026), who study connectedness among disaggregated CPI series for the United States, the United Kingdom and Japan since the late 1980s using the time-varying-parameter VAR (TVP-VAR) framework of Antonakakis et al. (2020). We make three contributions relative to that work. The first is *span*: we extend the sample back to the Great Inflation, so that the high inflation of the 1970s and the COVID-era high inflation of 2021–22 can be compared within a common empirical framework. The second is *scope*: alongside the six disaggregated CPI categories we include wages and inflation expectations, which lets us examine wage–price interactions and the inflation–expectations interactions; additionally, we examine the role price connectedness plays in the setting of monetary policy. The third is *methodological*: we replace the connectedness measures derived from the TVP-VAR with the pseudo-R^2 quantile connectedness approach of Shahzad et al. (2026), which better suits the question for two reasons. First, it measures connectedness at different points of the inflation distribution rather than only at the conditional mean or median, so the dependence structure in high-inflation states can differ from that in normal times.[1] Second, it separates *contemporaneous* co-movement from *lagged* propagation, letting us distinguish time series that move together within the period from those linked through delayed pass-through; this is important in distinguishing common shocks from lagged propagation from price series to price series (that is, spiral type behaviour). Throughout, we emphasise that connectedness measures statistical association and lead–lag dependence, not identified causal transmission; our claims concern how the dependence structure shifts across inflation states, not what causes what.

This paper tests six hypotheses. The first tests whether inflation becomes more broad-based when inflation is high:

H1 (CPI broadening). As inflation rises, connectedness among disaggregated CPI categories increases, particularly in the upper tail of the inflation distribution.

The second asks whether high-inflation states are also associated with stronger wage–price dependence:

H2 (Wage–price interaction). The connectedness between wages and prices is stronger during high-inflation states than during normal-inflation states.

[1] A rolling-window TVP-VAR could in principle track the two episodes through time, but it still measures connectedness at the centre of the conditional distribution within each window; it cannot isolate behaviour in the tail.

We then compare two major U.S. inflation episodes: the Great Inflation of the 1970s and the COVID-era inflation surge of 2021–22. Much of the conventional understanding of high-inflation dynamics is informed by the 1970s. The pandemic-era surge provides a modern episode, under a very different monetary policy regime, in which inflation became sufficiently high to revisit those price/wage and inflation dynamics in high inflation environments. The two episodes differed in important respects. The Great Inflation occurred in an economy more dependent on oil and a different monetary policy framework. The COVID-era inflation, differs from the Great Inflation, in that supply shock that affected many prices simultaneously; the pandemic-related supply constraints, in part owing to shipping disruptions was not just concentrated in oil-related phenomenon. These differences suggest that the two episodes may have broadened through different channels. This motivates the third hypothesis:

H3 (Episode mechanism). The COVID-era surge was characterised primarily by contemporaneous connectedness, consistent with a large common shock, whereas the Great Inflation was characterised more by lagged propagation across prices.

The differences in monetary policy frameworks between the two periods of high inflation motivate the fourth hypothesis. Specifically, we study if the adoption of inflation targeting and the associated anchoring of inflation expectations, alter the connectedness of inflation expectations and inflation:

H4 (Expectations channel). Inflation expectations were a stronger source of inflation propagation during the Great Inflation than during the COVID-era surge.

Finally, we consider whether the breadth of price changes shapes the conduct of monetary policy. As we noted earlier, there is an argument that monetary policy should respond to broad-based, core inflation whilst 'looking through' idiosyncratic movements in individual components (relative price changes).[2] All else equal, then, when price changes are highly connected across the subcomponents of the CPI, and as such are less likely to reflect idiosyncratic price changes, such inflation may be harder to look through, and monetary policy makers may respond to it more strongly:

H5 (Connectedness and monetary policy). When the subcomponents of the CPI are more highly connected, monetary policy responds more strongly to inflation.

Further, when realised and expected inflation aree more tightly connected at high levels of inflation, the risk of a self-reinforcing spiral rises: expectations may be behaving adaptively rather than remaining anchored. Policy makers may then respond more aggressively to forestall such a spiral. This motivates our final hypothesis:

H6 (Connectedness and monetary policy II). When inflation spills over more strongly into inflation expectations at high levels of inflation, monetary policy responds more strongly to inflation.

Our central finding is that the connectedness among prices changes with the level of inflation. Cross-category connectedness among the CPI subcomponents rises roughly threefold from the median to the high-inflation tail (from 11.8 to 35.2 per cent at $\tau = 0.9$), and the increase is accounted

[2]See Goodfriend and King, 1997, p.276

for increasingly by the *lagged* rather than the contemporaneous component (23.3 against 11.9 per cent at $\tau = 0.9$). We also find hetereogeneity in connectedness between the two periods of high inflation: the COVID-era surge chiefly through contemporaneous co-movement, consistent with a common supply shock, and the Great Inflation more through lagged propagation across categories. We find wages and prices are more tightly connected at the top end of the distribution, especially during the Great Inflation, Further, as expected during the Great Inflation we see strong evidence consistent with a wage price spiral lagged inflation spills over into wages and lagged inflation spills over into inflation. We also find interesting results regarding the connectedness between inflation and inflation expectations. Over the 1983–2019 period where it could be argued inflation expectations were well achonored, expectations transmit to inflation (net connectedness +29.1 at $\tau = 0.9$); but once either high-inflation episode enters the sample period the sign reverses, so expectations *follow* realised inflation, consistent with adaptive rather than forward-looking behaviour. Finally, we some suggestive evidence that monetary policy responds more strongly to inflation when CPI subcomponent price changes are more highly connected.

The remainder of the paper proceeds as follows. Section 2 describes the data; Section 3 sets out the method; and Section 4 reports the tests of the hypotheses.

2 Data

Hypotheses 1, 3 and 5 are to do with the connectedness between CPI subcomponents. Our study uses six monthly CPI subcomponent series; these series collectively cover all of the headline CPI without overlapping. Each is expressed as a month-on-month percentage change and are downloaded from the FRED website. The sample covers February 1967 to May 2026. The six subcomponents are:

- Core goods (excluding food)
- Core services (excluding shelter)
- Shelter
- Food
- Household energy
- Gasoline

There are two exceptions from the standard expenditure-class decomposition: energy is separated into gasoline and household energy services, and core services into shelter and core services excluding shelter. The shelter series is available from FRED, but the core-services-ex-shelter series, we construct as:

$$\%\Delta\text{CoreServ}^{\text{ex-shelter}} = \frac{\%\Delta\text{CoreServ} - w\,\%\Delta\text{Shelter}}{1-w}, \tag{1}$$

where $w = 0.60$ is estimated shelter's share of core services.[3]

We separate shelter from other core services because shelter inflation is driven by distinct dynamics and may behave differently from other services' prices. Rents are nominally rigid within tenancies: a large share of continuing tenants see no nominal rent change from one year to the next (Genesove, 2003), a stickiness that arises naturally when landlords facing search frictions prefer to retain existing tenants (Gallin & Verbrugge, 2019). Moreover, the CPI shelter component is measured from a slowly rotating panel dominated by continuing tenants, so measured shelter inflation responds to market (new-tenant) rents only with a lag of about a year (Adams et al., 2024; Bolhuis et al., 2022). Both features give shelter a smoother, more persistent inflation process than other core services, and a distinctive lagged relationship to the rest of the CPI — precisely the kind of heterogeneity our connectedness framework is designed to capture.

To test hypotheses 2, 4 and 6, we use three additional series, again from FRED. For the overall Consumer Price Index, we use the Consumer Price Index for All Urban Consumers, All Items, transformed into a month-on-month percentage change to obtain headline inflation.[4] Wage growth is measured using the average hourly earnings of production and nonsupervisory employees in the total private sector (`AHETPI`, dollars per hour, seasonally adjusted), expressed as a month-on-month percentage change. We use this series rather than the broader all employees measure because it is the longest-running monthly earnings series, beginning in 1964, and therefore spans the Great Inflation. A long span of data is needed for the for comparative analysis of wage and inflation connectedness between the Great Inflation and the COVID Inflation. The broader all-employees series begins only in 2006.

Finally, we use the University of Michigan median expected change in prices over the next twelve months (`MICH`, percent), available monthly from January 1978. The monthly series therefore misses the 1973–77 period, a key period of the Great Inflation. The Michigan measure is, however, available back to 1960 at a quarterly frequency, so we also run the expectations analysis at a quarterly frequency.

[3]The results are insensitive to this choice: connectedness and the directional structure are stable for $w \in [0.50, 0.65]$ (reported in the robustness analysis).

[4]We omit the overall CPI from the subcomponent system because it is constructed from those very subcomponents, so any connectedness found would in part reflect a mechanical accounting relationship rather than economic co-movement.

3 Methodology

3.1 Pseudo-quantile R^2 connectedness

Let $\boldsymbol{y}_t = (y_{1,t}, \ldots, y_{K,t})'$ collect the K time series we study. The precise membership of $\boldsymbol{y}_t$ depends on the hypothesis under study. For hypotheses 1, 3 and 5, $\boldsymbol{y}_t$ contains the month-on-month percentage change of the six CPI subcomponents. For hypotheses 4 and 5 it contains inflation expectations and headline inflation. For hypothesis 2 it contains wage growth and headline inflation.

For each series $y_{k,t}$ in turn within the system $\boldsymbol{y}_t$, we ask how much of its variation can be attributed to (i) variation of the remaining series in the system at time t, $\boldsymbol{y}_{-k,t}$, and (ii) the first p lags of all the time series in the whole system, $(\boldsymbol{y}'_{t-1}, \ldots, \boldsymbol{y}'_{t-p})'$. Channel (i) captures *contemporaneous* dependence: the extent to which variables move together within the month (or the quarter), while the second channel (ii) captures *lagged* dependence; that is, delayed propagation from one series to another over time. The ability to isolate and distinct between the two channels is what allows us to separate a common shock from sequential pass-through.

A natural starting point to decompose the two channels is the goodness of fit of a regression as measured by the R^2. Consider quantifying the connectedness between inflation (π) and inflation expectations (π^e) the following regression:

$$\pi_t = \alpha_0 + \alpha_1 \pi_t^e + \sum_{i=1}^{p} \beta_i \pi_{t-i}^e + \sum_{i=1}^{p} \gamma_i \pi_{t-i} + \epsilon_t.$$

The R^2 tells us how well collectively inflation expectations in time t and lagged inflation expectations are correlated with inflation, and how inflation is correlated with its own lags. The issue for the researcher wondering about the relative role of contemporaneous vs lagged inflation expectations in the inflation process is the lags are correlated with the contemporaneous value of inflation expectations, so the regression R^2 cannot be partitioned unambiguously into the portions that reflect the lagged and the contemporaneous contributions respectively. This problem is exacerbated when you move to more than two variables (like the six individual subcomponents of the CPI) and therefore have contemporaneous and lagged values of two or more independent variables in the regression; if these variables are correlated, the partitioning of the R^2 becomes even more fraught.

Genizi (1993) proposes an order-invariant decomposition of the regression R^2 that distributes it across correlated regressors through the symmetric (eigenvalue) square root of their correlation matrix. Let $\boldsymbol{R}_{xx}$ denote the correlation matrix of the regressors, with eigendecomposition $\boldsymbol{R}_{xx} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}'$, where $\boldsymbol{V}$ holds the eigenvectors and $\boldsymbol{\Lambda}$ the eigenvalues. Defining the symmetric square root

$\boldsymbol{C}$, the vector of decomposed R^2 shares is

$$\boldsymbol{R}_{xx} = \boldsymbol{V}\,\boldsymbol{\Lambda}\,\boldsymbol{V}' = \boldsymbol{C}\,\boldsymbol{C}', \tag{2}$$

$$\boldsymbol{C} = \boldsymbol{V}\,\boldsymbol{\Lambda}^{1/2}\,\boldsymbol{V}', \tag{3}$$

$$\boldsymbol{R}^{2,d} = \left[\,\boldsymbol{C}^2\,(\boldsymbol{C}^{-1}\boldsymbol{r}_{yx})^2\,\right]' \tag{4}$$

where $\boldsymbol{r}_{yx}$ is the vector of correlations between the dependent variable and each regressor. Each entry of $\boldsymbol{R}^{2,d}$ is the share of the regression R^2 owed uniquely to a single regressor, net of the correlation it shares with the others; the entries are non-negative and sum to the regression R^2.

Using the above decomposition with Pearson correlations as inputs, Balli et al. (2023) show how contemporaneous and lagged connectedness can be quantified at the centre of the distribution. Because, however, our interest is in how dependence behaves when inflation is high, we use the adaptation of Shahzad et al. (2026), which replaces the Pearson correlations with the quantile correlations of Choi and Shin (2022); this allows contemporaneous and lagged connectedness to be measured at any chosen quantile τ.

The quantile correlation of Choi and Shin (2022) between two variables x_1 and x_2 at quantile τ is

$$r_{1,2,\tau} = r_{2,1,\tau} = \operatorname{sign}\big(b_{x_1|x_2}(\tau)\big)\sqrt{b_{x_1|x_2}(\tau)\,b_{x_2|x_1}(\tau)}, \qquad 0 < \tau < 1, \tag{5}$$

where $b_{x_1|x_2}(\tau)$ and $b_{x_2|x_1}(\tau)$ are the slope coefficients from the τ-quantile regressions of x_1 on x_2 and of x_2 on x_1, respectively. At $\tau = 0.5$ this is a median-based analogue of the Pearson correlation, whereas for τ in the tails it captures dependence specific to large or small realisations.

The specifics of the Shahzad et al. (2026) measure are as follows. The quantile correlation of Choi and Shin (2022) between two variables, $r_{1,2,\tau}$, is substituted for $\boldsymbol{R}_{xx}$ and $\boldsymbol{r}_{yx}$ in equations 2 and 4. For each time series in turn, equation 4 then returns a vector $\boldsymbol{R}^{2,d}_{\tau}$ that allocates the regression's overall pseudo-R^2 across its regressors at that τ.[5] The first $K-1$ entries record the contemporaneous contributions of the other series; the remaining entries record the lagged contributions of all series in the system.[6]

For a system of K variables, applying the decomposition with the quantile correlation for each of the K equations in that system and stacking the results yields the connectedness matrix

$$\boldsymbol{R}^{2,d}_{\tau} = \begin{bmatrix} \boldsymbol{R}^{2,d}_{\tau,1} \\ \vdots \\ \boldsymbol{R}^{2,d}_{\tau,K} \end{bmatrix} = \left[\,\boldsymbol{R}^{2,d}_{lag=0,\tau} \;\middle|\; \boldsymbol{R}^{2,d}_{lag=1,\tau} \;\middle|\; \cdots \;\middle|\; \boldsymbol{R}^{2,d}_{lag=p,\tau}\,\right], \tag{6}$$

[5]Quantile regressions do *not* have an R^2 in the conventional (least-squares) sense, hence the term "pseudo".

[6]A quantile-correlation matrix estimated from a finite sample need not be positive semi-definite; a difficulty that becomes more pronounced in the tails ($\tau = 0.9$ in our data) and in short rolling windows. As per Shahzad et al. (2026). we apply the Schäfer and Strimmer (2005) shrinkage procedure

in which row k indexes the dependent variable.[7]

The total connectedness index at quantile τ, TCI_τ, is the average across the K series of the off-diagonal $R^{2,d}$ shares and splits into a contemporaneous and a lagged component,

$$\mathrm{TCI}_\tau = \mathrm{TCI}_\tau^C + \mathrm{TCI}_\tau^L, \tag{7}$$

with

$$\mathrm{TCI}_\tau^C = \frac{1}{K}\sum_{k=1}^{K}\sum_{j\neq k}\boldsymbol{R}^{2,d}_{lag=0,\tau,k,j}, \qquad \mathrm{TCI}_\tau^L = \frac{1}{K}\sum_{k=1}^{K}\sum_{j\neq k}\sum_{l=1}^{p}\boldsymbol{R}^{2,d}_{lag=l,\tau,k,j}, \tag{8}$$

where $\boldsymbol{R}^{2,d}_{lag=l,\tau,k,j}$ is the share of series k's variation contributed by series j at lag l. Throughout, connectedness measures are reported as percentages (shares of the R^2 multiplied by 100). Equivalently, TCI_τ is the average of the directional FROM — or, identically, TO — measures defined below.

Directional measures in a similar spirit to those put forward by Diebold and Yilmaz (2012) can also be calculated. Both are defined for a single series k: its FROM measure is the row-k sum of the connectedness matrix and its TO measure the column-k sum. The total connectedness *received* by series k (its FROM measure) sums the shares contributed by all *other* series, at every lag, along row k:

$$\mathrm{FROM}_{k,\tau} = \sum_{j\neq k}\boldsymbol{R}^{2,d}_{lag=0,\tau,k,j} + \sum_{j\neq k}\sum_{l=1}^{p}\boldsymbol{R}^{2,d}_{lag=l,\tau,k,j}. \tag{9}$$

The total connectedness *transmitted* by series k to the other series (its TO measure) sums the shares it contributes, at every lag, down column k:

$$\mathrm{TO}_{k,\tau} = \sum_{j\neq k}\boldsymbol{R}^{2,d}_{lag=0,\tau,j,k} + \sum_{j\neq k}\sum_{l=1}^{p}\boldsymbol{R}^{2,d}_{lag=l,\tau,j,k}. \tag{10}$$

The net directional connectedness of series k,

$$\mathrm{NET}_{k,\tau} = \mathrm{TO}_{k,\tau} - \mathrm{FROM}_{k,\tau}, \tag{11}$$

is positive when k is, on balance, a transmitter of connectedness and negative when it is a net receiver.

We prefer this measure of quantile connectednes to other approaches to quantile connectedness, such as that of Ando et al. (2022). Those measures propagate the non-zero-mean residuals from quantile regressions through a generalised forecast-error variance decomposition that treats them as zero-mean shocks. As Shahzad et al. (2026) show, this inflates measured connectedness in both

[7] So that the equations align, the kth position of $\boldsymbol{R}^{2,d}_{\tau,k}$ is set to zero, since $y_{k,t}$ is not regressed on its own contemporaneous value.

tails.

4 Results

4.1 H1: inflation broadens in the high tail

Our first hypothesis is **H1 (CPI broadening)**: *as inflation rises, connectedness among disaggregated CPI categories increases, particularly in the upper tail of the inflation distribution.* Figure 1 reports the overall connectedness at different levels of τ over our whole sample period on a monthly frequency (the black line). Further, the figure shows the decomposition into contemporaneous (the blue line) and lagged (the red line) connectedness. Overall connectedness rises about threefold from the median to the high tail (from 11.8 to 35.2 at monthly frequency); both the contemporaneous and lagged components increase, but slightly more of the rise comes from the lagged component (23.3 vs 11.9 at $\tau = 0.9$). These estimates of connectedness are based off estimating the system with two lags; longer lag lengths are not possible as there is a risk the quartile-correlation matrix is no longer positive semi definite.[8] This means the connectedness is being measured over a relatively short span of time: two months. We also run the system on a quarterly frequency. The idea being that two lags now capture a longer span of time i.e. six months; this allows more time for prices to spillover into other prices. Table 1 reports connectedness measured at a monthly (Panel A) and quarterly (Panel B) frequency for selected quartiles. The table indicates that key conclusions from the monthly analysis hold at a quarterly frequency: high-inflation states are characterised by increased connectedness and, in particular, increased lagged connectedness. Based on this evidence, we conclude hypothesis one is strongly supported.

[Figure 1 about here.]

[Table 1 about here.]

4.2 H2: Wages and inflation

The second hypothesis states that connectedness between wages and prices is stronger during high-inflation states than during normal-inflation states. Panel A of Table 2 shows that the connectedness between wages and the overall CPI rises with the quantile, and that at higher quantiles the lagged inflation–wage association strengthens; this pattern holds at quarterly frequency (Panel C). This provides strong support for Hypothesis 3.

Where the monthly and quarterly data differ is the monthly data suggest at high inflation, lagged spillovers are stronger from inflation to wage growth than from lagged wage growth to the CPI; the quarterly data point to the opposite conclusion. We read this as a lag-window effect

[8] As we use shrinkage procedures (Schäfer & Strimmer, 2005) to keep the matrix positive semi definite, the real issue is the shrinkage required to keep it positive semi-definite grows large enough to distort the resulting connectedness estimates.

rather than a contradiction. Wage-setting adjusts at multi-month horizons — annual reviews and contract renegotiations — so within the two-month window spanned by the monthly specification, only the faster price-to-wage pass-through (recent inflation feeding into wage demands and cost-of-living adjustments) has time to register. The quarterly specification's two lags span roughly six months, long enough for the slower wages-to-prices cost channel to appear, and it is precisely this channel that then dominates. The same sensitivity of measured direction to the lag window arises for goods-to-services sequencing in Appendix B, which is why we report both frequencies throughout.

To understand how wage and inflation dynamics changed during the two inflation episodes in our sample (the Great Inflation and the COVID inflation), we conduct a 'one in/ one out' exercise; see Panel B. In the Great-Inflation-inclusive subperiod (1964-2019): wage–price connectedness is much stronger and the spillovers are in both directions; this is consistent with a reinforcing spiral. For example, at $\tau = 0.9$, the 1964–2019 sample has spillovers from inflation to wages at 15.5 and 14.9 in the opposite direction i.e. roughly even. Examining the results from the sample including COVID, but excluding the Great Inflation (1983-2026), we make two observations. The first is the spillovers are relatively stronger from inflation to wages (3.0) than in the other direction (1.3). The second is level of connectedness is much lower. In part, this reflects that the level of inflation was much higher during the Great Inflation, so a given τ is associated with higher inflation. It may also reflect that with better anchored inflation expectations, wage price spirals are less likely.[9]

[Table 2 about here.]

4.3 H3: The two high inflation episodes were different in terms of how price changes propagated at a subcompoment level

Table 3 reports the marginal change in high-tail ($\tau = 0.9$) connectedness changes when each episode is added to the 1983–2019 sample. That is, estimating the model with the Great Inflation included (but COVID dropped) and then with COVID included (but the Great Inflation dropped). Both contemporaneous and lagged connectedness rise by more in the Great Inflation, consistent with its higher inflation level. More tellingly, the Great-Inflation increase comes mainly through the *lagged* component, whereas the COVID increase is almost entirely *contemporaneous*, its lagged marginal is in fact negative. The COVID results are thus consistent with a common shock (pandemic supply constraints and shipping disruption), while the Great-Inflation pattern with relatively lagged propogation playing a more important role, supporting H3.

At a quarterly frequency (Panel B) the picture at $\tau = 0.7$ is, if anything, sharper than at monthly frequency: the Great Inflation adds 25.6 points of connectedness, 73 per cent of it through the lagged

[9]These two channels — the mechanical one (a given τ corresponds to a higher inflation *level* in the Great Inflation than post-1983) and the structural one (anchoring) — can be partially separated by comparing the episodes at the *same* inflation level rather than the same τ. Appendix E does this for the six-CPI system: even level-matched, the Great Inflation's marginal broadening far exceeds COVID's, so the difference is not purely a level artefact.

channel, while COVID's marginal is essentially zero. At $\tau = 0.9$, however, the quarterly episode marginals are less informative, for two reinforcing reasons. First, the calm-era base is already high and lagged-dominated (the 1983–2019 core has overall connectedness of 76.1 at $\tau = 0.9$, of which 51.4 is lagged), leaving little room for an episode to add. Second, aggregation shrinks the sample: the COVID window contributes only 26 quarters, so the far tail of the quarterly distribution rests on a handful of observations. We therefore read the monthly results, and the quarterly results at $\tau = 0.7$, as the primary evidence.

[Table 3 about here.]

4.4 H4: Inflation expectations and headline inflation

Table 4 reports the directional connectedness of inflation expectations and headline inflation, together with the net directional connectedness of expectations (NET = TO − FROM; positive values mean expectations lead, negative that they follow).[10] Note this is carried out on a quarterly frequency so we can study a sample that includes the Great Inflation; monthly expectations data only become available in January 1978 (a monthly version of this analysis, on the post-1978 sample, is reported in Appendix C and is consistent with the quarterly results). Pooled over the full sample, inflation expectations are net transmitters in the upper tail. Taking $\tau = 0.9$ as example, contemporaneous plus lagged transmission from inflation to inflation expectations is (19.5+33.6) 53.1; it is (26.2+37.3) 63.5 in the other direction. If one excludes the two large inflation episodes in our sample and focuses on the 1983-2019 period, the stronger connectedness from inflation expectations to inflation (rather than vice-versa) is still evident. Examining the two high inflation episodes in isolation, however, we see in the Great Inflation the lagged inflation→expectations term dominates at higher τ (reaching 50.7 at $\tau = 0.9$, against 27.5 in the reverse direction). This is consistent with the adaptive channel: recent inflation spillovers into expectations. COVID shows a lower level of lagged connectedness from inflation expectations to inflation; this is consistent with better-anchored inflation expectations.

Unlike Table 3, the episode rows in Table 4 are estimated on each episode in isolation rather than as one-in/one-out marginals against the 1983–2019 core; isolation is the natural design here because the pre-1983 survey is what makes the Great Inflation observable at all in this system. For comparability, Appendix D reports the one-in/one-out construction as well; it tells a consistent story — the Great-Inflation marginal is large and overwhelmingly lagged at both $\tau = 0.7$ and $\tau = 0.9$, while COVID's marginal is essentially zero at $\tau = 0.7$ (its positive $\tau = 0.9$ marginal rests on the short 26-quarter window and should be read with the same caution as Panel B of Table 3).

[10] Inflation here is quarter-on-quarter growth in the CPI. There is an argument we should instead use annual (four-quarter) growth, to match the one-year horizon of the survey question. At a quarterly frequency, however, overlapping four-quarter changes share three of their four quarters of price data with the adjacent observations, mechanically inducing a moving-average component in the series; this smoothness inflates measured *lagged* connectedness in both directions and blurs precisely the contemporaneous-versus-lagged and directional distinctions the table is built on. Re-running the table with four-quarter inflation leaves the net lead/follow pattern across samples intact, though the size of the Great-Inflation lagged asymmetry is sensitive to the choice; these results are available from the authors on request.

[Table 4 about here.]

4.5 H5 and H6: connectedness and the conduct of monetary policy

We motivated hypotheses five and six by noting that monetary policy makers might be more concerned when inflation is more board-based. To test H5 and H6 we proceed in three steps. First, for each connectedness system we compute the pseudo-quantile R^2 connectedness measure over rolling windows, using both a 120-month window and evaluating it at $\tau = 0.5$, 0.7 and 0.9 so as to characterise connectedness at the centre as well as in the upper tail of the distribution.[11] This yields, for each quantile, a monthly series of total connectedness, C_t^τ.

Second, we classify each month into a high- or low-connectedness regime. Rather than imposing an ad hoc threshold, we let the data determine the regimes by fitting the two-state Markov-switching model of Hamilton (1989) to C_t^τ. Specifically, we model connectedness as a first-order autoregressive process whose intercept, autoregressive coefficient and error variance all switch between two latent states, $s_t \in \{\text{low}, \text{high}\}$, that evolve as a first-order Markov chain. The AR(1) specification captures the persistence of the connectedness series (in part, brought about owing to the rolling window construction) whilst allowing both its level and its volatility to differ across regimes. We estimate the model by maximum likelihood, recover the smoothed probability that each month belongs to each state, and assign every month to its most probable state; the state with the higher mean connectedness is labelled the 'high' regime, so that $\text{High}_t = 1$ for months in that state and zero otherwise. Note because C_t^τ differs for different τs, the high and low connectedness regimes will be different for different τs.

Third, we estimate an inertial Taylor rule augmented with a regime interaction:

$$i_t = \alpha + \rho\, i_{t-1} + \beta_\pi \left(\pi_t - \pi^*\right) + \delta \left(\text{High}_t \cdot \left(\pi_t - \pi^*\right)\right) + \beta_x\, x_t + \varepsilon_t, \tag{12}$$

where i_t is the Wu–Xia (Wu & Xia, 2016) shadow policy rate, π_t is twelve-month CPI inflation, $\pi^* = 2$ per cent is the assumed inflation target, and x_t is the unemployment gap.[12] The coefficient ρ captures interest-rate smoothing (Clarida et al., 2000, p.151-152); one motivation for this specification is that under uncertainty central banks only partially adjust to the interest rate warranted based on current economic conditions. In equation 12, β_π estimates the baseline response of monetary policy to the inflation gap, and β_x the response to unemployment. The object of interest is δ: it measures the *additional* monetary response to inflation in the high-connectedness regime. If H5 and H6 are true, then $\delta > 0$: monetary policy responds more strongly to inflation when price changes (H5) or inflation and inflation expectations (H6) are more highly connected. All inference is based

[11] We choose this window to balance two competing requirements. Too short a window means connectedness (particularly at higher τs) depends on a small number of months. Too long a window means that the connectedness measure is not dynamic enough; that is, it evolves too slowly to capture policy-makers' perceptions of the broadness of inflation.

[12] The unemployment gap is the difference between the civilian unemployment rate and the Congressional Budget Office's estimate of the natural (non-cyclical) rate of unemployment, both obtained from FRED (series `UNRATE` and `NROU`). Because the CBO series is published quarterly, we linearly interpolate it to monthly frequency.

on Newey–West HAC standard errors.

Table 5 reports estimates of equation (12) in which the high- and low-connectedness regimes are identified from the connectedness of the six CPI subcomponents, measured over a 120-month rolling window. We re-estimate the rule over several sample start dates because the monetary-policy reaction function is known to have shifted over time, with different Federal Reserve chairs placing different weight on inflation (Clarida et al., 2000); estimating over later subsamples guards against attributing such governor-preference changes in the rule to the connectedness regime.

Two readings of the evidence presented in Table 5 are possible. On the first reading, the connectedness state dependence of monetary policy is real, but recent: it emerges only in the samples beginning in 1993 and 2003; this is consistent with modern central banks, operating under explicit or implicit inflation targets. being more attentive to the breadth of price changes than their predecessors. That the effect appears only at $\tau = 0.9$ reinforces this reading, suggesting that policymakers respond more strongly to inflation only once connectedness is concentrated in the upper tail. That is, it is when CPI subcomponents are experiencing large price increases that policy makers start to care about spillovers. This interpretation has some plausiblity.

A second, more sceptical, reading is that the two significant coefficients are false positives. The interaction δ is estimated over twelve quantile–sample combinations, so at conventional levels one or two spurious rejections of the null hypothesis are to be expected by chance alone; that one of the two is significant only at the 10 per cent level lends this reading further weight. Balancing these two readings of the data, we therefore treat the result as no more than suggestive support for Hypothesis H5.

[Table 5 about here.]

Table 6 reports the Taylor-rule estimates for the six CPI subcomponents when the connectedness regime is based on connecedness measured at quarterly frequency. The advantage of this approach is that subcomponent changes have a longer span of time (two quarters as opposed to two months) to propagate through the system. The evidence for connectedness state dependent monetary policy is notably stronger than under the monthly measure. Of the eight estimates of the state dependence δ at $\tau = 0.5$ and $\tau = 0.7$ (four sample start dates at each quantile), six are positive and statistically significant at the 10 per cent level or better (most at 5 per cent or better), indicating a stronger inflation response in high-connectedness months. At $\tau = 0.9$, by contrast, only one of the four estimates is statistically significant.

[Table 6 about here.]

Table 7 reports the Taylor rule estimates when the connectedness regime is identified from the inflation–expectations system. Panel A shows the monthly analysis, where inflation is measured using the month-on-month percentage change in the CPI; the monthly expectations survey only begins in 1978, so this panel's earliest sample starts there. Panel B shows the quarterly analysis, where inflation is measured using the quarter-on-quarter percentage change and connectedness is measured at a quarterly frequency, so that its two lags span roughly six months. At monthly frequency (Panel A) there is no evidence of connectedness-state-dependent policy: the only two δ estimates that reach the 10 per cent level carry opposite signs, the pattern one expects from chance. At quarterly frequency (Panel B) the evidence is somewhat stronger — δ is positive and significant at the 1 per cent level at $\tau = 0.5$ in the 1993 and 2003 samples, and at the 5 per cent level at $\tau = 0.7$ in the 2003 sample — echoing the six-CPI finding that the quarterly (longer-propagation-horizon) connectedness measure is where state dependence appears, and again concentrated in the later, inflation-targeting-era samples. The Extended column, which exploits the native quarterly Michigan survey to push the sample back to 1969 and so spans the Great Inflation and the Volcker disinflation, shows no significant interaction at any quantile. As with H5, then, three significant cells out of twelve — none in the upper tail, and none in the long sample — fall well short of decisive: either the connectedness state dependence of policy toward expectations is a recent, inflation-targeting-era phenomenon, or the scattered significance is chance. We read the evidence for H6 as suggestive at best.[13]

[Table 7 about here.]

A Connectedness matrices

Tables 8–13 report the full pairwise connectedness structure among the six CPI subcomponents at $\tau = 0.9$ ($n_{\text{lag}} = 2$), separately for the contemporaneous and lagged blocks, for the full sample and the two subperiods (1967–1982 and 1983–2026). In each matrix, cell (i, j) is the share of row i's variation attributed to column j; FROM is the row sum (received), TO the column sum (transmitted), and NET = TO−FROM. The contemporaneous block is close to symmetric (small NET), whereas the lagged block carries the directional structure — shelter and core services are the net lagged transmitters. Gasoline is weakly connected to the other categories: its variation is dominated by energy-market (crude-oil) movements that lie outside the CPI system, so it sits at the periphery of the network.

[Table 8 about here.]

[Table 9 about here.]

[Table 10 about here.]

[Table 11 about here.]

[13] An earlier version of this analysis paired the expectations series with twelve-month (annual) CPI inflation, matched to the survey's one-year horizon. For the reasons discussed above (the moving-average component that overlapping annual changes induce), we use the short-horizon measure throughout; the matched-measure results, which show a similar scattering of significance, are available from the authors on request.

[Table 12 about here.]

[Table 13 about here.]

B Goods-to-services sequencing: a lag-window illustration

A frequently cited feature of the 2021–22 surge is that goods inflation led services inflation. Our framework recovers this, but only once the measurement window is long enough — which is itself instructive about connectedness measurement. Table 14 reports, on the post-1983 sample at $\tau = 0.9$, the net lagged connectedness from core goods to core services. Under the monthly two-lag dependence (Genizi) metric the sign is slightly negative (-1.2): a two-month window is too short to span a multi-month goods-then-services sequence. The LASSO predictive metric, which admits longer lags, recovers the ordering ($+1.3$ at two lags, $+5.9$ at six, where core goods becomes a net transmitter). Table 15 confirms it at quarterly frequency, where a two-lag block spans six months: the lagged goods$\rightarrow$services term then exceeds the reverse at $\tau = 0.5$ (9.9 vs 4.4) and $\tau = 0.7$ (11.2 vs 7.1). The point is methodological — lag-window length materially affects what connectedness detects, which motivates the quarterly analysis used elsewhere — rather than a new fact about the episode.

[Table 14 about here.]

[Table 15 about here.]

C Monthly robustness for the inflation–expectations system

Table 4 in the main text is estimated at a quarterly frequency, because only the quarterly Michigan survey reaches back before the Great Inflation. Table 16 reports the monthly analogue on the post-1978 sample, pairing the monthly Michigan series with month-on-month CPI inflation. The qualitative pattern matches the quarterly results: expectations lead in the calm 1983–2019 core (NET $+24.1$ at $\tau = 0.9$) and follow during COVID, and in the portion of the Great Inflation the monthly survey covers (1978–82) the lagged terms dominate the contemporaneous ones. The 1978–82 sample contains only 60 months, so its estimates — particularly at $\tau = 0.9$ — carry wide uncertainty and should be read as corroborative rather than free-standing evidence.

[Table 16 about here.]

D One-in/one-out episode marginals for the expectations system

The episode rows of Table 4 are estimated on each episode in isolation. For comparability with the six-CPI episode analysis (Table 3), Table 17 reports the same one-in/one-out construction for the inflation–expectations system: each episode's marginal contribution to total connectedness relative to the calm 1983–2019 core. The Great-Inflation marginal is large and overwhelmingly lagged at both quantiles (83 and 105 per cent lagged shares at $\tau = 0.7$ and 0.9), consistent with the adaptive channel operating through recent realised inflation. COVID's marginal is essentially zero at $\tau = 0.7$; its positive $\tau = 0.9$ marginal rests on the 26 quarters the window adds and warrants the same caution as the quarterly episode results in the main text.

[Table 17 about here.]

E Comparing episodes at the same inflation level

Comparing the Great Inflation and COVID at the same τ compares different inflation levels: within the Great-Inflation sample, annualised monthly headline inflation at $\tau = 0.9$ is 13.1 per cent, against 8.7 per cent within the COVID sample. Some of the difference in measured broadening between the episodes could therefore be mechanical — a given τ simply "means" more inflation in the 1970s. Table 18 addresses this by matching *levels* rather than quantiles, in two ways. Panels A and B fix a common reference level (each episode's $\tau = 0.9$ headline rate in turn) and re-estimate the episode marginals at the τ that corresponds to that level within each episode; Panel C fixes a common absolute threshold of 5 per cent annualised. The conclusion is the same throughout: even at the same inflation level, the Great Inflation's marginal broadening substantially exceeds COVID's (for example, $+12.2$ against $+1.1$ at the common 8.7 per cent level, and $+5.6$ against $+0.7$ at the common 5 per cent threshold), and remains predominantly lagged. The episode difference is therefore not a level artefact, consistent with the anchoring interpretation advanced in the main text.

[Table 18 about here.]

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Table 1 H1 — Inflation broadens into the high tail

Panel A. Monthly total connectedness by quantile

τ	Overall	Contemporaneous	Lagged
0.1	5.5	2.8	2.8
0.3	7.5	2.7	4.8
0.5	11.8	4.7	7.2
0.7	19.7	7.1	12.6
0.9	35.2	11.9	23.3

Panel B. Quarterly total connectedness by quantile

τ	Overall	Contemporaneous	Lagged
0.1	16.9	8.3	8.6
0.3	17.6	6.8	10.8
0.5	27.7	10.7	17.1
0.7	55.7	21.6	34.1
0.9	86.8	33.8	52.9

Notes: Pseudo-quantile R^2 (Genizi/nearPD) connectedness (%) among the six CPI subcomponents; Overall $=$ Contemporaneous $+$ Lagged, $n_{\text{lag}} = 2$. Panel A is monthly 1967–2026 (broadening ratio $\text{TCI}(0.9)/\text{TCI}(0.5) = 2.98\times$); Panel B quarterly 1967Q2–2026Q2 (ratio $3.13\times$). Both frequencies show the tail rise, increasingly driven by the lagged component. Point estimates; bootstrap confidence intervals for the six-CPI system are to be added.

Table 2 H2 — Wage–price connectedness is stronger when inflation is high

Panel A. Two-variable wage–CPI system (monthly, full sample), by quantile

	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.9$
Wage–CPI total connectedness (TCI)	10.8	16.6	21.6
lagged Wages→CPI	5.0	7.7	11.5
lagged CPI→Wages	6.2	9.2	16.1

Panel B. Quarterly (full sample), by quantile

	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.9$
Wage–CPI total connectedness (TCI)	25.7	32.6	53.4
lagged Wages→CPI	14.6	19.3	35.1
lagged CPI→Wages	13.0	18.5	23.8

Panel C. By subperiod (monthly), by quantile

	$\tau = 0.5$	$\tau = 0.7$	$\tau = 0.9$
1964–2019 (with Great Inflation)			
Wage–CPI total connectedness (TCI)	13.3	18.4	25.2
lagged Wages→CPI	6.0	9.1	15.5
lagged CPI→Wages	5.9	9.0	14.9
1983–2026 (post-Great-Inflation)			
Wage–CPI total connectedness (TCI)	1.2	1.8	3.6
lagged Wages→CPI	0.2	0.2	1.3
lagged CPI→Wages	0.9	1.7	3.0

Panel D. 1964–2019 (with Great Inflation), level-matched τ (monthly)

	Matched to τ=0.7	Matched to τ=0.9
Wage–CPI total connectedness (TCI)	14.7	22.1
lagged Wages→CPI	7.2	13.2
lagged CPI→Wages	7.4	13.3

Notes: Two-variable system of wage growth (AHETPI) and overall CPI (CPIAUCSL); pseudo-quantile R^2 connectedness (%), Genizi estimator, $n_{\text{lag}} = 2$. Wage–price connectedness rises sharply into the high-inflation tail at all frequencies (Panels A–B), and is far stronger in the Great-Inflation-inclusive subperiod (1964–2019) than post-1983 (Panel C). In the high tail the lagged components dominate. Quarterly ($n_{\text{lag}} = 2$) spans a six-month horizon. Panel D addresses the fact that comparing the two subperiods at the same τ compares different inflation *levels*: it reports the 1964–2019 connectedness at the quantiles ($\tau = 0.62$ and 0.78) whose month-on-month CPI-inflation level matches that of the post-1983 sample at $\tau = 0.7$ and $\tau = 0.9$ (about 4.0 and 6.3 per cent annualised). Even level-matched, 1964–2019 connectedness (14.7 and 22.1) remains far above the post-1983 values in Panel C (1.8 and 3.6), so the gap is not a level artefact; the level-matched spillovers also stay roughly symmetric, the balanced feedback characteristic of a spiral.

Table 3 H3 — COVID broadened contemporaneously; the Great Inflation through lagged propagation

Panel A. Monthly: marginal $\Delta TCI(\tau)$ vs. the 1983–2019 core

Episode	ΔOverall	ΔContemp.	ΔLagged	Lagged %
$\tau = 0.7$				
GreatInflation (1967-82)	+12.3	+4.4	+7.9	64.4
COVID (2020-26)	-0.1	+0.0	-0.2	139.6
$\tau = 0.9$				
GreatInflation (1967-82)	+14.3	+5.0	+9.3	64.9
COVID (2020-26)	+1.1	+1.9	-0.8	-74.7

Panel B. Quarterly: marginal $\Delta TCI(\tau)$ vs. the 1983–2019 core

Episode	ΔOverall	ΔContemp.	ΔLagged	Lagged %
$\tau = 0.7$				
GreatInflation (1967-82)	+25.6	+6.9	+18.6	72.9
COVID (2020-26)	-0.1	-0.4	+0.3	-405.9
$\tau = 0.9$				
GreatInflation (1967-82)	+3.2	+3.8	-0.6	-17.8
COVID (2020-26)	-19.9	-5.0	-14.9	74.9

Notes: Each episode's marginal contribution to connectedness at $\tau \in \{0.7, 0.9\}$ relative to the calm 1983–2019 core (nested subtraction), split into contemporaneous and lagged; six-CPI system, Panel A monthly, Panel B quarterly. At monthly frequency the Great Inflation broadened mainly through the *lagged* (propagation) channel at both quantiles, while COVID broadened contemporaneously (its lagged marginal is negative at $\tau = 0.9$ and essentially zero at $\tau = 0.7$). At quarterly frequency the same contrast is sharp at $\tau = 0.7$; at $\tau = 0.9$ the calm-era base is already high and lagged-dominated, and the COVID window contributes only 26 quarters, so the episode marginals there are less informative.

Table 4 H4 — Contemporaneous and lagged connectedness between inflation and expectations

		Inflation → Expectations		Expectations → Inflation		
Sample	τ	Contemp.	Lagged	Contemp.	Lagged	NET
Full (1960–2026)	0.5	24.2	23.7	27.4	25.4	+4.8
	0.7	20.9	28.5	27.7	35.4	+13.7
	0.9	19.5	33.6	26.2	37.3	+10.5
Core (1983–2019)	0.5	17.9	8.8	22.8	6.8	+3.0
	0.7	14.9	6.2	21.6	8.2	+8.8
	0.9	21.3	11.9	27.5	34.7	+29.1
Great Inflation (1967–82)	0.5	28.4	27.7	27.6	29.0	+0.5
	0.7	22.4	28.9	23.0	26.2	-2.2
	0.9	22.5	50.7	25.5	27.5	-20.2
COVID (2020–26)	0.5	0.3	6.1	0.4	0.5	-5.5
	0.7	1.6	11.9	1.7	7.7	-4.1
	0.9	0.8	7.8	1.3	0.6	-6.7

Notes: Pairwise pseudo-quantile R^2 connectedness (%) in a two-variable quarterly system of Michigan inflation expectations (`MICH_QTR`, from 1960) and overall CPI (CPIAUCSL), $n_{\text{lag}} = 2$ (a six-month lag horizon). "Inflation → Expectations" is the share of *expectations'* variation explained by CPI, split into contemporaneous and lagged; "Expectations → Inflation" is the reverse. NET is the net directional connectedness of expectations (TO − FROM); > 0 means expectations lead (transmit), < 0 that they follow. The *lagged* Inflation → Expectations term (the adaptive channel) dominates in the high tail of both episodes — most strongly in the Great Inflation (50.7 at $\tau = 0.9$) — and expectations are net followers there, whereas in the calm 1983–2019 period expectations lead. Dependence, not identified causation.

Table 5 Subcomponent connectedness

	Earliest (1977)	1983	1993	2003
$\tau = 0.5$				
Intercept	0.16^{**} (2.1)	0.10^{*} (1.7)	0.04 (0.9)	0.04 (1.1)
ρ (i_{t-1})	0.95^{***} (54.3)	0.96^{***} (62.3)	0.99^{***} (96.5)	0.98^{***} (95.4)
β_π ($\pi - \pi^*$)	0.06^{**} (2.3)	0.02 (1.0)	-0.03 (-0.6)	0.02 (1.1)
$\Delta\beta_\pi^{\text{High}}$	0.01 (0.3)	0.03 (1.4)	0.07 (1.3)	0.03 (1.4)
β_x (ugap)	-0.04^{**} (-2.4)	-0.03^{*} (-1.8)	-0.02 (-1.5)	-0.03^{**} (-2.3)
n	590	518	398	278
high-conn. share	0.40	0.33	0.63	0.15
long-run β_π (low/high)	1.19/1.31	0.53/1.41	-2.91/3.92	0.94/2.52
$\tau = 0.7$				
Intercept	0.16^{**} (2.2)	0.11^{*} (1.8)	0.04 (0.9)	0.04 (1.0)
ρ (i_{t-1})	0.95^{***} (54.7)	0.97^{***} (61.6)	0.99^{***} (98.4)	0.98^{***} (97.5)
β_π ($\pi - \pi^*$)	0.03 (1.1)	-0.01 (-0.4)	-0.02 (-0.5)	0.02 (1.3)
$\Delta\beta_\pi^{\text{High}}$	0.04 (1.5)	0.06 (1.6)	0.06 (1.2)	0.03 (1.3)
β_x (ugap)	-0.04^{**} (-2.5)	-0.03^{*} (-1.8)	-0.02 (-1.5)	-0.03^{**} (-2.2)
n	590	518	398	278
high-conn. share	0.62	0.62	0.58	0.16
long-run β_π (low/high)	0.59/1.31	-0.41/1.29	-2.20/3.67	1.34/2.81
$\tau = 0.9$				
Intercept	0.17^{**} (2.2)	0.10 (1.6)	0.03 (0.9)	0.04 (1.3)
ρ (i_{t-1})	0.95^{***} (55.4)	0.97^{***} (61.9)	0.99^{***} (116.0)	0.98^{***} (104.4)
β_π ($\pi - \pi^*$)	0.04 (1.2)	0.03 (1.2)	0.01 (0.4)	0.01 (0.7)
$\Delta\beta_\pi^{\text{High}}$	0.03 (1.1)	0.02 (0.6)	0.04^{**} (2.0)	0.04^{*} (1.8)
β_x (ugap)	-0.04^{**} (-2.3)	-0.03^{*} (-1.7)	-0.02^{*} (-1.8)	-0.03^{**} (-2.6)
n	590	518	398	278
high-conn. share	0.45	0.37	0.26	0.31
long-run β_π (low/high)	0.67/1.33	0.82/1.32	0.45/3.52	0.57/2.42

Notes: Estimates of equation (12). Columns are sample start dates; each panel a quantile τ. Newey–West HAC t-statistics in parentheses; *, **, *** denote significance at 10, 5 and 1 per cent. $\Delta\beta_\pi^{\text{High}}$ is the interaction δ; the final row reports the implied long-run response $\beta_\pi/(1-\rho)$ in the low and high regimes.

Table 6 Subcomponent connectedness: Quarterly

	Earliest (1977)	1983	1993	2003
$\tau = 0.5$				
Intercept	0.17^{**} (2.2)	0.12^{*} (1.9)	0.03 (0.9)	0.04 (1.2)
ρ (i_{t-1})	0.95^{***} (52.4)	0.96^{***} (61.7)	0.99^{***} (132.4)	0.98^{***} (112.7)
β_π ($\pi - \pi^*$)	0.05^{**} (2.3)	0.01 (0.4)	0.01 (0.4)	0.01 (0.6)
$\Delta\beta_\pi^{\text{High}}$	0.03 (1.5)	0.06^{***} (2.7)	0.06^{***} (3.2)	0.05^{***} (3.0)
β_x (ugap)	-0.05^{**} (-2.6)	-0.04^{**} (-2.4)	-0.02^{**} (-2.3)	-0.03^{***} (-3.0)
n (months)	589	517	397	277
high-conn. share	0.32	0.28	0.13	0.19
long-run β_π (low/high)	0.91/1.42	0.14/1.55	0.38/4.41	0.39/2.85
$\tau = 0.7$				
Intercept	0.16^{**} (2.1)	0.11^{*} (1.8)	0.02 (0.5)	0.03 (0.8)
ρ (i_{t-1})	0.95^{***} (54.2)	0.96^{***} (61.1)	0.99^{***} (127.8)	0.98^{***} (109.5)
β_π ($\pi - \pi^*$)	0.02 (1.3)	0.01 (0.7)	0.00 (0.1)	0.01 (0.5)
$\Delta\beta_\pi^{\text{High}}$	0.05^{**} (2.2)	0.03 (1.4)	0.05^{**} (2.2)	0.04^{*} (1.8)
β_x (ugap)	-0.05^{***} (-2.6)	-0.04^{**} (-2.4)	-0.02^{**} (-2.0)	-0.02^{**} (-2.1)
n (months)	589	517	397	277
high-conn. share	0.42	0.34	0.20	0.31
long-run β_π (low/high)	0.46/1.35	0.37/1.24	0.14/5.14	0.60/3.06
$\tau = 0.9$				
Intercept	0.16^{**} (2.4)	0.11^{*} (1.7)	0.03 (0.7)	0.03 (1.0)
ρ (i_{t-1})	0.95^{***} (57.9)	0.96^{***} (59.3)	0.99^{***} (111.0)	0.98^{***} (105.0)
β_π ($\pi - \pi^*$)	0.07^{***} (3.5)	0.02 (0.7)	0.01 (0.6)	0.01 (0.5)
$\Delta\beta_\pi^{\text{High}}$	-0.03 (-1.4)	0.03 (1.3)	0.03 (1.3)	0.04^{*} (1.7)
β_x (ugap)	-0.04^{**} (-2.3)	-0.04^{**} (-2.0)	-0.02^{*} (-1.7)	-0.02^{**} (-2.3)
n (months)	589	517	397	277
high-conn. share	0.40	0.44	0.39	0.36
long-run β_π (low/high)	1.41/0.89	0.40/1.25	0.83/3.13	0.60/2.86

Notes: Estimates of equation (12). Columns are sample start dates; each panel a quantile τ. Newey–West HAC t-statistics in parentheses; *, **, *** denote significance at 10, 5 and 1 per cent. $\Delta\beta_\pi^{\text{High}}$ is the interaction δ; the final row reports the implied long-run response $\beta_\pi/(1-\rho)$ in the low and high regimes.

Table 7 Expectations connectedness and the policy reaction function: 120-month window

Panel A. Monthly: MICH and month-on-month CPI inflation

	Earliest (1987)	1993	2003
$\tau = 0.5$			
Intercept	0.05 (1.4)	0.02 (0.6)	0.03 (0.8)
ρ (i_{t-1})	0.98*** (113.9)	0.99*** (110.0)	0.99*** (99.1)
β_π ($\pi - \pi^*$)	0.00 (0.1)	−0.00 (-0.1)	0.04*** (3.3)
$\Delta\beta_\pi^{\text{High}}$	0.03 (1.1)	0.05* (1.8)	−0.05* (-1.8)
β_x (ugap)	−0.03** (-2.5)	−0.02 (-1.6)	−0.03** (-2.0)
n	458	397	277
high-conn. share	0.46	0.42	0.55
long-run β_π (low/high)	0.13/1.59	−0.29/4.54	3.34/ − 0.35
$\tau = 0.7$			
Intercept	0.05 (1.4)	0.04 (0.9)	0.04 (1.1)
ρ (i_{t-1})	0.98*** (113.7)	0.98*** (109.1)	0.98*** (91.9)
β_π ($\pi - \pi^*$)	0.04 (1.6)	0.03 (1.1)	0.03 (1.4)
$\Delta\beta_\pi^{\text{High}}$	−0.01 (-0.5)	0.01 (0.2)	0.00 (0.1)
β_x (ugap)	−0.03** (-2.4)	−0.02* (-1.9)	−0.03** (-2.2)
n	458	397	277
high-conn. share	0.47	0.39	0.30
long-run β_π (low/high)	1.71/1.02	1.74/2.22	1.60/1.74
$\tau = 0.9$			
Intercept	0.05 (1.4)	0.04 (0.9)	0.05 (1.2)
ρ (i_{t-1})	0.98*** (117.2)	0.98*** (105.9)	0.98*** (93.3)
β_π ($\pi - \pi^*$)	0.03 (1.6)	0.03 (1.3)	0.04* (1.7)
$\Delta\beta_\pi^{\text{High}}$	−0.00 (-0.2)	0.02 (1.0)	−0.00 (-0.2)
β_x (ugap)	−0.03*** (-2.6)	−0.02* (-2.0)	−0.03** (-2.2)
n	458	397	277
high-conn. share	0.40	0.35	0.36
long-run β_π (low/high)	1.34/1.15	1.57/2.57	1.73/1.58

Panel B. Quarterly: MICH and quarter-on-quarter CPI inflation

	Extended† (1969)	Earliest (1987)	1993	2003
$\tau = 0.5$				
Intercept	0.18*** (2.8)	0.05 (1.5)	0.03 (1.0)	0.04 (1.3)
ρ (i_{t-1})	0.95*** (61.6)	0.98*** (116.2)	0.98*** (131.1)	0.98*** (114.4)
β_π ($\pi - \pi^*$)	0.06** (2.0)	0.01 (0.7)	−0.00 (-0.0)	0.01 (0.4)
$\Delta\beta_\pi^{\text{High}}$	−0.01 (-0.5)	0.02 (1.0)	0.06*** (3.3)	0.06*** (3.1)
β_x (ugap)	−0.06*** (-2.9)	−0.03*** (-2.7)	−0.02** (-2.5)	−0.03*** (-3.1)
n (months)	674	461	398	278
high-conn. share	0.30	0.40	0.27	0.21
long-run β_π (low/high)	1.06/0.87	0.51/1.58	−0.02/3.86	0.29/2.86
$\tau = 0.7$				
Intercept	0.16** (2.4)	0.06 (1.5)	0.03 (0.9)	0.01 (0.5)
ρ (i_{t-1})	0.95*** (62.9)	0.98*** (120.3)	0.99*** (116.6)	0.98*** (108.5)
β_π ($\pi - \pi^*$)	0.03 (1.3)	0.03* (1.8)	0.02 (1.1)	−0.01 (-0.5)
$\Delta\beta_\pi^{\text{High}}$	0.04 (1.4)	−0.01 (-0.4)	0.02 (0.9)	0.06** (2.1)
β_x (ugap)	−0.05** (-2.4)	−0.03** (-2.6)	−0.02** (-2.1)	−0.02** (-2.0)
n (months)	674	461	398	278
high-conn. share	0.49	0.54	0.18	0.65
long-run β_π (low/high)	0.60/1.44	1.49/1.11	1.66/3.29	−0.60/2.70
$\tau = 0.9$				
Intercept	0.16** (2.4)	0.05 (1.3)	0.03 (0.8)	0.04 (1.2)
ρ (i_{t-1})	0.95*** (67.6)	0.98*** (115.8)	0.99*** (118.4)	0.98*** (87.8)
β_π ($\pi - \pi^*$)	0.03 (1.3)	0.00 (0.1)	0.02 (0.6)	0.04** (2.2)
$\Delta\beta_\pi^{\text{High}}$	0.03 (0.6)	0.03 (1.0)	0.03 (0.8)	0.00 (0.0)
β_x (ugap)	−0.05*** (-3.1)	−0.03*** (-2.8)	−0.02** (-2.2)	−0.03** (-2.2)
n (months)	674	461	398	278
high-conn. share	0.57	0.42	0.33	0.27
long-run β_π (low/high)	0.71/1.35	0.10/1.71	1.22/3.14	1.62/1.63

Notes: Estimates of equation (12) in which the connectedness regime is identified from the inflation–expectations system over a 120-month rolling window. Panel A pairs the Michigan one-year inflation expectation with month-on-month CPI inflation, with connectedness measured at monthly frequency (the monthly survey begins in 1978). Panel B pairs it with quarter-on-quarter CPI inflation, with connectedness measured at quarterly frequency and each month assigned the regime of its quarter. Columns are sample start dates; within each panel the three blocks are the quantiles $\tau = 0.5$, 0.7 and 0.9. † The Extended column in Panel B uses the native quarterly Michigan survey (available from 1960, so the 40-quarter rolling window delivers connectedness from 1969); the remaining Panel B columns use the monthly survey (from 1978) aggregated to quarters, and are retained for comparability with Panel A. Newey–West HAC t-statistics in parentheses; *, **, *** denote significance at 10, 5 and 1 per cent. $\Delta\beta_\pi^{\text{High}}$ is the interaction δ; the final row of each block reports the implied long-run response $\beta_\pi/(1-\rho)$ in the low and high regimes.

Table 8 Contemporaneous connectedness among CPI subcomponents (full sample, 1967–2026, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	—	0.2	1.6	4.6	5.4	1.2	12.9
Gasoline	0.4	—	4.9	0.1	0.1	0.1	5.6
HH energy	1.6	4.4	—	1.8	1.0	1.2	9.9
Core goods	4.6	0.0	1.8	—	6.4	2.5	15.3
Shelter	4.9	0.0	0.5	6.3	—	5.8	17.5
Core svc. ex-sh.	0.8	0.1	1.2	2.3	5.7	—	10.1
TO	12.2	4.7	10.0	15.2	18.5	10.8	
NET	-0.7	-0.9	+0.0	-0.2	+1.0	+0.7	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\mathrm{lag}} = 2$. Cell (i, j) is the share of row i's variation attributed to column j. FROM = row sum (received); TO = column sum (transmitted); NET = TO−FROM ($>$ 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO.

Table 9 Lagged connectedness among CPI subcomponents (full sample, 1967–2026, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	23.5	0.3	5.5	6.6	13.1	4.6	30.1
Gasoline	1.4	11.3	1.4	0.3	0.3	0.3	3.6
HH energy	2.8	2.8	10.2	2.6	1.9	6.2	16.3
Core goods	6.8	0.4	3.1	29.3	5.6	9.9	25.8
Shelter	12.0	0.1	1.3	6.5	26.6	12.8	32.7
Core svc. ex-sh.	3.5	0.9	3.6	7.1	16.2	15.4	31.4
TO	26.5	4.5	14.9	23.1	37.1	33.8	
NET	-3.6	+0.9	-1.4	-2.8	+4.4	+2.4	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i's variation attributed to column j. FROM = row sum (received); TO = column sum (transmitted); NET = TO−FROM ($>$ 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO (and shown for the lagged block as the own-lag/persistence term).

Table 10 Contemporaneous connectedness among CPI subcomponents (1967–1982, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	—	4.9	1.9	8.5	2.9	1.3	19.6
Gasoline	3.6	—	6.9	4.0	4.7	7.9	27.0
HH energy	1.5	18.9	—	3.1	1.6	2.8	28.1
Core goods	4.3	11.5	2.3	—	4.6	4.8	27.5
Shelter	2.2	16.2	2.0	4.8	—	10.5	35.8
Core svc. ex-sh.	2.9	16.6	1.9	4.1	5.4	—	31.0
TO	14.6	68.2	15.0	24.6	19.3	27.3	
NET	-5.0	+41.2	-13.1	-2.9	-16.5	-3.7	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i's variation attributed to column j. FROM = row sum (received); TO = column sum (transmitted); NET = TO−FROM (> 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO.

Table 11 Lagged connectedness among CPI subcomponents (1967–1982, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	31.5	2.1	23.2	9.4	3.8	6.0	44.4
Gasoline	22.3	24.4	10.9	4.3	8.4	2.8	48.6
HH energy	15.8	8.1	15.9	19.7	4.8	7.3	55.8
Core goods	6.9	4.8	17.4	34.7	5.0	3.4	37.6
Shelter	15.5	6.3	2.8	20.4	16.5	2.5	47.5
Core svc. ex-sh.	16.2	9.5	3.6	18.6	5.3	15.5	53.3
TO	76.7	30.9	57.9	72.4	27.4	21.9	
NET	+32.3	-17.7	+2.1	+34.9	-20.1	-31.4	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i's variation attributed to column j. FROM = row sum (received); TO = column sum (transmitted); NET = TO−FROM ($>$ 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO (and shown for the lagged block as the own-lag/persistence term).

Table 12 Contemporaneous connectedness among CPI subcomponents (1983–2026, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	—	0.2	6.3	0.3	2.2	1.0	9.9
Gasoline	0.4	—	4.0	0.1	3.5	0.3	8.2
HH energy	6.5	3.8	—	1.1	0.2	0.4	12.0
Core goods	0.3	0.1	1.0	—	0.4	4.1	5.8
Shelter	2.2	2.9	0.2	0.4	—	2.8	8.5
Core svc. ex-sh.	1.0	0.2	0.4	4.2	2.8	—	8.6
TO	10.4	7.2	11.8	6.0	9.1	8.4	
NET	+0.5	-1.0	-0.2	+0.2	+0.6	-0.1	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i's variation attributed to column j. FROM = row sum (received); TO = column sum (transmitted); NET = TO−FROM ($>$ 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO.

Table 13 Lagged connectedness among CPI subcomponents (1983–2026, $\tau = 0.9$)

	Food	Gas.	HH en.	Core gds	Shelter	Core svc.	FROM
Food	14.6	1.6	9.2	2.6	4.8	0.7	18.9
Gasoline	3.6	10.3	3.0	0.0	0.8	0.6	8.0
HH energy	4.6	3.8	10.8	0.8	0.8	1.2	11.2
Core goods	2.6	1.6	1.4	30.7	0.7	3.0	9.3
Shelter	7.5	0.5	0.5	0.5	19.9	8.3	17.4
Core svc. ex-sh.	3.1	8.1	3.1	2.1	8.8	5.7	25.1
TO	21.4	15.6	17.3	6.0	15.8	13.9	
NET	+2.4	+7.6	+6.1	-3.3	-1.6	-11.3	

Notes: Pseudo-quantile R^2 connectedness (%) among the six CPI subcomponents at $\tau = 0.9$, $n_{\text{lag}} = 2$. Cell (i, j) is the share of row i's variation attributed to column j. FROM = row sum (received); TO = column sum (transmitted); NET = TO−FROM ($>$ 0 net transmitter). Diagonal (own) terms are excluded from FROM/TO (and shown for the lagged block as the own-lag/persistence term).

Table 14 Goods-to-services sequencing under alternative metrics and lag lengths

Metric	Goods→services (lagged, net)	Core goods NET
Genizi dependence (n_{lag}=2)	-1.2	-3.4
LASSO predictive (n_{lag}=2)	+1.3	+3.4
LASSO predictive (n_{lag}=6)	+5.9	+12.9

Notes: "Goods→services (lagged, net)" is lagged connectedness from core goods to core services (shelter + core services ex-shelter) minus the reverse, at $\tau = 0.9$ on the ex-Great-Inflation sample (1983–2026); > 0 means goods lead. The two-lag Genizi dependence metric cannot see a multi-month sequence and tilts negative; the LASSO predictive metric confirms the sequencing, increasingly at longer lags. Predictive ordering, not identified causation.

Table 15 Goods-to-services: contemporaneous and lagged connectedness (quarterly)

	Goods $\rightarrow$ Services		Services $\rightarrow$ Goods	
τ	Contemp.	Lagged	Contemp.	Lagged
0.5	3.3	9.9	3.4	4.4
0.7	8.5	11.2	8.7	7.1
0.9	13.1	18.4	15.2	18.3

Notes: Pairwise pseudo-quantile R^2 connectedness (%) between core goods and core services (shelter + core services ex-shelter) on the ex-Great-Inflation quarterly sample (1983Q1–2026Q2, 174 quarters), $n_{\text{lag}} = 2$ (a six-month lag horizon). "Goods $\rightarrow$ Services" is the share of services' variation explained by goods, split into contemporaneous and lagged; "Services $\rightarrow$ Goods" is the reverse. The *lagged* Goods $\rightarrow$ Services term is the "goods lead" propagation channel; it exceeds the lagged Services $\rightarrow$ Goods term at every quantile (by 4.0 at $\tau = 0.7$), consistent with goods leading services. Six months is the horizon the monthly two-lag metric could not span. Dependence, not identified causation.

Table 16 Monthly robustness — connectedness between inflation and expectations

Sample	τ	Inflation $\rightarrow$ Expectations		Expectations $\rightarrow$ Inflation		
		Contemp.	Lagged	Contemp.	Lagged	NET
Full (1978–2026)	0.5	5.6	12.6	6.4	8.0	-3.9
	0.7	9.2	16.1	12.9	20.3	+7.9
	0.9	15.2	30.3	19.6	36.4	+10.4
Late Great Inflation (1978–82)	0.5	16.9	32.5	20.1	23.5	-5.8
	0.7	12.5	27.9	13.9	22.1	-4.4
	0.9	5.1	8.9	4.7	15.5	+6.2
Core (1983–2019)	0.5	5.6	7.0	8.0	3.2	-1.4
	0.7	5.8	8.5	11.2	7.3	+4.2
	0.9	11.4	11.4	20.7	26.2	+24.1
COVID (2020–26)	0.5	1.7	11.6	2.8	3.8	-6.7
	0.7	4.6	8.7	5.4	3.6	-4.3
	0.9	3.2	5.5	3.4	4.2	-1.1

Notes: Monthly analogue of Table 4: pairwise pseudo-quantile R^2 connectedness (%) in a two-variable system of the Michigan one-year inflation expectation (FRED `MICH`, available monthly from January 1978) and month-on-month CPI inflation (CPIAUCSL), $n_{\text{lag}} = 2$ (a two-month lag horizon). NET is the net directional connectedness of expectations (TO $-$ FROM); > 0 means expectations lead, < 0 that they follow. Because the monthly survey begins in 1978, the Great-Inflation sample covers only its final years (1978–82, 60 months) and should be read with caution. Dependence, not identified causation.

Table 17 Episode marginals for the inflation–expectations system (one-in/one-out)

Episode	ΔOverall	ΔContemp.	ΔLagged	Lagged %
$\tau = 0.7$				
Great Inflation (pre-1983 in)	+28.5	+4.8	+23.7	83.1
COVID (2020– in)	-0.1	-4.5	+4.4	-4533.3
$\tau = 0.9$				
Great Inflation (pre-1983 in)	+14.4	-0.8	+15.1	105.2
COVID (2020– in)	+12.2	+3.6	+8.7	70.8

Notes: One-in/one-out episode marginals for the quarterly two-variable system of Michigan inflation expectations (`MICH_QTR`) and quarter-on-quarter CPI inflation, constructed exactly as in Table 3: each row is total connectedness estimated with that episode included (and the other excluded) minus the calm 1983–2019 core, split into contemporaneous and lagged; $n_{\text{lag}} = 2$ (a six-month lag horizon). The pre-1983 sample runs from 1960, so the "Great Inflation" marginal includes the 1960s run-up. Dependence, not identified causation.

Table 18 Level-matched episode comparison: same inflation rate, episode-specific τ

Episode	Level (% ann.)	τ used	ΔOverall	ΔContemp.	ΔLagged
Panel A. Common level: COVID $\tau = 0.9$ headline rate as reference					
Great Inflation	8.70	0.693	+12.2	+4.4	+7.9
COVID	8.70	0.900	+1.1	+1.9	-0.8
Panel B. Common level: Great-Inflation $\tau = 0.9$ headline rate as reference					
Great Inflation	13.08	0.900	+14.3	+5.0	+9.3
COVID	13.08	0.950†	+1.3	+5.3	-3.9
Panel C. Common absolute threshold: 5% annualised headline inflation					
Great Inflation	5.00	0.349	+5.6	+1.9	+3.6
COVID	5.00	0.645	+0.7	+0.6	+0.1

Notes: Comparing the two episodes at the same τ compares different inflation levels, because their inflation distributions differ. Here the comparison is level-matched: within each episode, τ is chosen so that the corresponding quantile of annualised month-on-month headline CPI inflation equals the common reference level shown. Δ columns are the episode's one-in/one-out marginal connectedness relative to the 1983–2019 core (as in Table 3), evaluated at that episode-specific τ; six-CPI system, monthly, $n_{\text{lag}} = 2$. † The implied τ lies outside $[0.05, 0.95]$ (the reference level sits at or beyond the edge of that episode's inflation distribution) and is clamped to the boundary; the comparison at that reference should be read as one-sided.

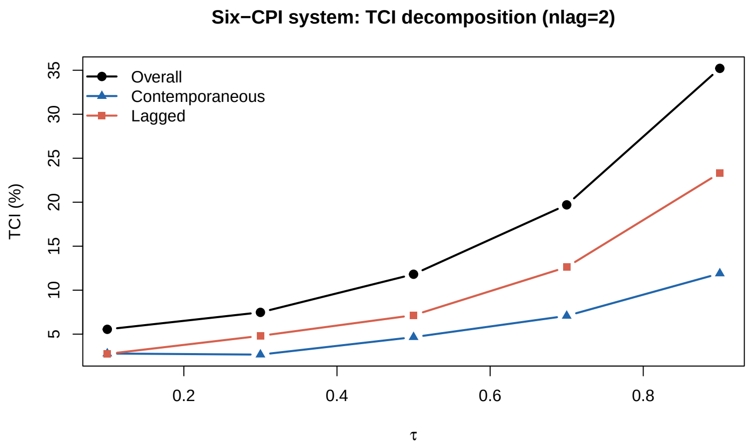

Fig. 1 Overall connectedness (black) and its decomposition into contemporaneous (blue) and lagged (red) components, by inflation quantile τ (monthly, $n_{\text{lag}} = 2$).